Limits of Work Extraction under Local Thermal Operations

A Numerical Study of Ergotropic Advantage and Correlation Dynamics

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Outline

1. Motivation & Problem Statement
2. Background: Ergotropy & LTO
3. Numerical Methodology
4. Results
5. Conclusion

Thermal Operations & Ergotropic Advantage

Thermal Operations (TO)

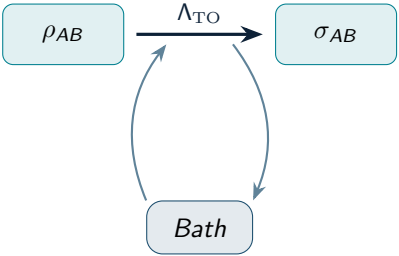
A **Thermal Operation** on a system ρ couples it to a thermal bath $\gamma_\beta = e^{-\beta H_R}/Z$ and applies a **global** energy-conserving unitary U:

$$\Lambda_{\mathrm{TO}}(\rho) = \mathrm{Tr}_R\left[U\left(\rho \otimes \gamma_\beta\right)U^\dagger\right]$$

with $[U,\, H_S + H_R] = 0$.

Ergotropy as Work Measure

For a state ρ with Hamiltonian H, the **ergotropy** $\mathcal{R}(\rho)$ is the maximum work extractable by any unitary.



$$\mathcal{R}(\sigma_{AB}) - \mathcal{R}(\rho_{AB}) > 0$$

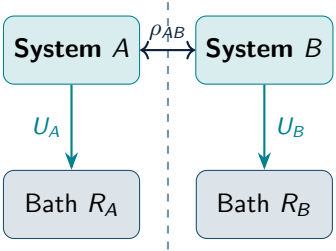
for some ρ_{AB}, U

Motivation & Core Question

Core Question

Can quantum correlations (eg, entanglement) be converted into **extractable work** using **purely local** thermal operations?

- **Global Thermal Ops (TO):** Collective manipulation of system + bath
- **Local Thermal Ops (LTO):** Each subsystem interacts *only* with its own reservoir
- The locality constraint raises a fundamental thermodynamic question.



Conjecture to Test

Ergotropy is a **monotone** under LTO for finite dimensional systems: $\Delta\mathcal{R} \leq 0$

Ergotropy & the LTO Framework

Ergotropy

Maximum work extractable via unitary evolution:

$$\mathcal{R}(\rho, H) := \mathrm{Tr}(\rho H) - \min_U \mathrm{Tr}(U\rho U^\dagger H)$$

Equivalently, using the **passive state** ρ_p:

$$\mathcal{R} = \mathrm{Tr}(\rho H) - \mathrm{Tr}(\rho_p H)$$

Global $\mathcal{R}_G$: unitary on full $d_A d_B$-space
Local $\mathcal{R}_L$: local unitaries $U_A \otimes U_B$
Ergotropic gap: $\delta\mathcal{R} = \mathcal{R}_G - \mathcal{R}_L$ detects quantum correlations.

LTO: Three-Stage Protocol

1. **Dilation**

$$\rho_{\mathrm{tot}} = \rho_{AB} \otimes \gamma_{R_A} \otimes \gamma_{R_B}$$

2. **Local Interaction** via energy-conserving $U = U_A \otimes U_B$:

$$[U_i,\, H_{S_i} + H_{R_i}] = 0$$

3. **Reduction** (trace out baths):

$$\sigma_{AB} = \mathrm{Tr}_{R_A,R_B}\left[(U)\,\rho_{\mathrm{tot}}\,(U)^\dagger\right]$$

Numerical Investigations

1. Scaling with Quantum Resources

Track $\Delta\mathcal{R}_G$ and $\Delta\mathcal{R}_L$ against:

- Mutual Information $I(A{:}B)$
- Negativity $\mathcal{N}$ (entanglement)
- Quantum Discord Q
- Classical Correlation C

2. Dynamics of the Ergotropic Gap

Analyze $\delta\mathcal{R} = \mathcal{R}_G - \mathcal{R}_L$ (known correlation detector): Does LTO dissipate shared resources?

3. System Configurations

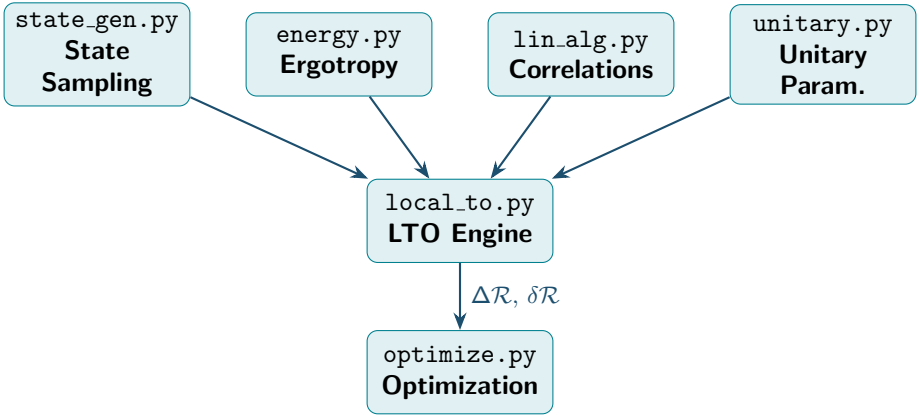
Test across four configurations:

- **(i)** Qubit – Qubit (Non-degenerate)
- **(ii)** Qubit – Qubit (Degenerate)
- **(iii)** Qubit – Harmonic
- **(iv)** Harmonic – Harmonic

4. Numerical Verification of Monotonicity

Show $\max(\Delta\mathcal{R}) \leq 0$ across a vast ensemble of states and optimized local unitaries $\Rightarrow$ numerical grounding for the LTO monotonicity conjecture.

Numerical Methodology



State Classes Generated: Product · Separable · Pure Entangled (Haar + Schmidt) · Mixed Entangled · Werner · Harmonic (Hilbert-Schmidt)

Correlation Measures: Mutual Information I · Negativity $\mathcal{N}$ · Classical Correlations C · Quantum Discord Q

Quantum State Sampling

Product & Separable

$\rho_A \otimes \rho_B$

Separable:

$\sum_i p_i \, \rho_{A,i} \otimes \rho_{B,i}$

Kronecker product of random local DMs; p_i normalized weights

Pure Entangled

Eg. : $|\psi\rangle = \sqrt{\alpha}|00\rangle + \sqrt{1-\alpha}|11\rangle$

Haar-random $|\psi\rangle$ - sampled from Normal Distribution, filtered $\mathcal{N}(\rho) > 0$; Schmidt α swept

Mixed Entangled

Convex mix of pure entangled states

Random weights; varies purity & entanglement

Werner

$\rho_W = p|\Phi^+\rangle\langle\Phi^+| + \frac{1-p}{d^2}\mathbb{I}$

p swept: probes entangled $\rightarrow$ separable transition via Bell fraction

Harmonic (HO)

Hilbert-Schmidt uniform mixed states

Haar state on d^2-dim space; partial trace over d-dim env; truncated Fock basis

All states validated: $\rho \geq 0$, $\mathrm{Tr}(\rho) = 1$. Ensemble size: $N = 200 - 500$.
Entanglement filtered via Peres-Horodecki (ρ^{T_B} eigenvalues).

Computing Ergotropy: Spectral Reordering

Passive Energy via Spectral Sort

$$E_p = \sum_{j=1}^{d} \lambda_j^{\downarrow} \, \varepsilon_j^{\uparrow}$$

- $\lambda_j^{\downarrow}$: eigenvalues of ρ in **descending** order
- $\varepsilon_j^{\uparrow}$: eigenvalues of H in **ascending** order
- Pairs largest populations with lowest energies

Implementation:

- `numpy.linalg.eigvalsh` for spectrum
- Stable argsort for reordering

Global vs. Local Ergotropy

Global $\mathcal{R}_G$:

- Sort on $d_A d_B$-dim space
- $H_{\text{tot}} = H_A \otimes I + I \otimes H_B$

Local $\mathcal{R}_L$:

- $\rho_A = \mathrm{Tr}_B(\rho_{AB}), \quad \rho_B = \mathrm{Tr}_A(\rho_{AB})$
- Partial traces via `einsum`
- Passive energies computed *independently*

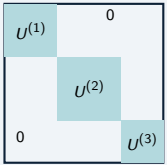
$$\boxed{\delta\mathcal{R} = \mathcal{R}_G - \mathcal{R}_L \geq 0}$$

Gap is an entanglement witness.

Energy-Conserving Unitary Parameterization

Block-Diagonal Structure

Constraint $[U_i, H_{S_i} + H_{R_i}] = 0$ forces U_i to be **block-diagonal** in the energy eigenbasis.



Each block $k \times k$ corresponds to a *degenerate energy subspace*.

Exponential Map Parameterization

Each block parameterized via:

$$U_k(\vec{\theta}) = \exp\big(iH(\vec{\theta})\big), \quad H^\dagger = H$$

Parameter vector $\vec{\theta} \in \mathbb{R}^{k^2}$:

- First k entries: **real diagonal** of H
- Remaining pairs: **Re/Im** of upper triangle
- Conjugate symmetry enforced $\Rightarrow H_{ji} = H_{ij}^*$

Validation: $\|[U_i, H_i]\|, \|U_i U_i^\dagger - I\| < 10^{-10}$

Optimization Strategies

Type 1 - Random Sampling

- Sample N_s initial states from each class
- For each state, apply N_u **Haar-random** energy-conserving unitaries.
- Identifies state and transformations *susceptible* to ergotropic improvement.

Sample N_s states → Apply N_u Haar unitaries → Record max $\Delta\mathcal{R}$

Type 2 - Nelder-Mead Refinement

- Promising states from Stage 1 passed to **local optimizer**
- `scipy.optimize.minimize (method='Nelder-Mead')`
- Objective: maximize $\Delta\mathcal{R}(\vec{\theta})$ over unitary parameters
- Multi-restart wrapper prevents convergence to **local minima**

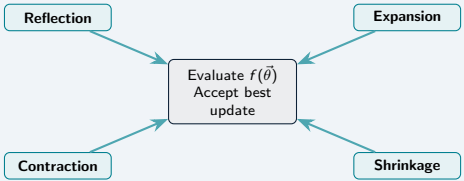
Convergence Criterion

$\max(\Delta\mathcal{R}) \leq 10^{-8}$ across all setups $\Rightarrow$ numerical verification of LTO monotonicity

Nelder-Mead: Algorithm & Restart Mechanism

The Simplex Method

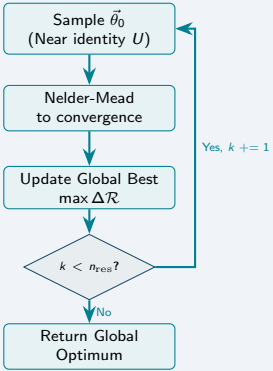
A **derivative-free** optimizer operating on $n+1$ vertices in $\mathbb{R}^n$.
At each iteration, the **worst vertex** $\vec{\theta}_w$ is replaced via:



Why here? Ergotropy involves spectral sorting, making it **non-differentiable** at eigenvalue crossings. Nelder-Mead avoids failed ∇f calculations.

Multi-Restart Mechanism

To avoid **local minima**, the restart loop ensures global manifold coverage:



LTO Simulation Engine

Index Reordering for Tensor Products

Joint space is ordered (S_A, S_B, R_A, R_B). Must **transpose** to (S_A, R_A, S_B, R_B) to apply $U_A \otimes U_B$ as a Kronecker product:

$$U_{\text{tot}} = U_A \otimes U_B$$

This avoids entangling cross-subsystem terms.

Validation Checks ($< 10^{-10}$ tol.)

- **Fixed point:** Gibbs state $\gamma_A \otimes \gamma_B$ unchanged
- **Bath invariance:** Local baths unaffected post-interaction
- **Trace conservation:** $\mathrm{Tr}(\sigma_{AB}) = 1$
- **Free energy monotonicity:** $F_i(\sigma_i) \leq F_i(\rho_i)$

Harmonic Extension

Truncated Hilbert space, dimension d. Density matrix ops scale as $\mathcal{O}(d^4)$. Single Nelder-Mead restart $\approx$ **30 min** wall-clock. Mutual information replaces Negativity as correlation marker.

Full Workflow Summary

1. **Initialization:** Set ω_1, ω_2, β_A, β_B, bath dimensions, harmonic dimension d, etc.
2. **Global Resource Mapping:** Batch of N states (e.g. pure entangled); apply fixed/optimized LTO; observe $\Delta\mathcal{R}$, $\delta\mathcal{R}$ vs. initial correlations
3. **State-Specific Unitary Sweeps:** Fix single Haar-random state; apply large ensemble of energy-conserving unitaries; track $\Delta\mathcal{R}$ vs. ΔI, $\Delta\mathcal{N}$, etc.
4. **Optimization & Monotonicity Testing:**
   - Monte Carlo global search over N_s states $\times$ N_u unitaries
   - Nelder-Mead refinement with n_{restarts} random initializations
   - Verify $\max(\Delta\mathcal{R}) \leq 10^{-8}$ as numerical threshold

Result 1: Non-Degenerate Qubit-Qubit Setup

Setup:

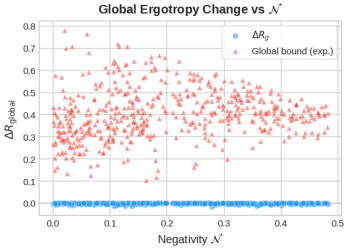
- $\omega_1 = 1.0$, $\omega_2 = 2.0$; $\beta_A = 1.0$, $\beta_B = 2.0$
- 500 initial states: random
- Monte Carlo optimization

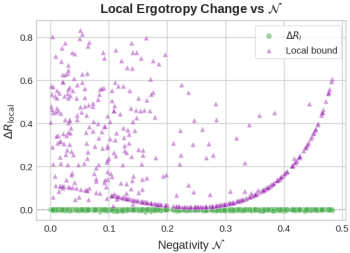
Key Findings:

- All states lie **at or below** $\Delta\mathcal{R} = 0$
- Even at max Negativity ($\mathcal{N} = 0.5$): no $\mathcal{R}$ gain
- Best LTO unitary $\equiv$ **Identity**
- Analytical bound $\Delta\mathcal{R} \leq 0$ confirmed numerically

Analytical Grounding

Non-degenerate LTO $\equiv$ **phase-damping channel**
$\Rightarrow \lambda(\sigma) \prec \lambda(\rho)$ (majorization)
$\Rightarrow$ passive energy can only *increase*





Result 2: Degenerate Qubit-Qubit Setup

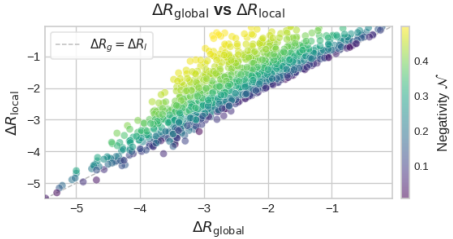
Setup:

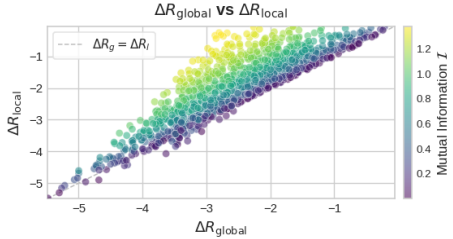
- Matched frequencies: $\omega_{S_1} = \omega_{B_1} = 1.0$, $\omega_{S_2} = \omega_{B_2} = 2.0$
- Non-trivial energy-conserving transformations *within* degenerate subspaces
- Focus on **pure entangled states**

Two Numerical Sweeps:

1. **Unitary Ensemble** (fixed state): tracks $\Delta\mathcal{R}$ vs. $\mathcal{N}$, I, Q, C
2. **State Ensemble** (500 states): global vs. local change per state

Outcome: Despite greater operational freedom, $\Delta\mathcal{R} \leq 0$ in *all* cases.



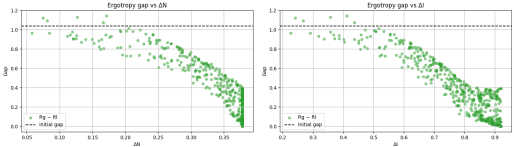


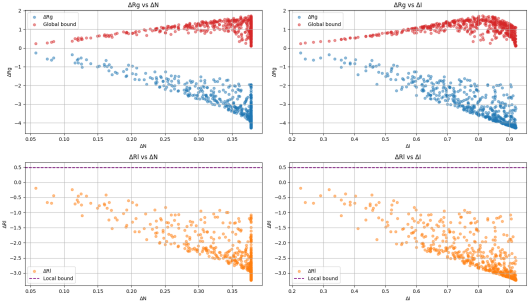
Result 3: Correlation Dissipation Dynamics

Ergotropic Gap Decay

The gap $\delta\mathcal{R} = \mathcal{R}_G - \mathcal{R}_L$ **dissipates toward zero** for larger $\Delta\mathcal{N}$:

- LTO eliminates non-local work potential
- $|\Delta\mathcal{R}_L| < |\Delta\mathcal{R}_G|$ for high-correlation states
- Global work loss $>$ local work loss $\Rightarrow$ shared resources "locked"





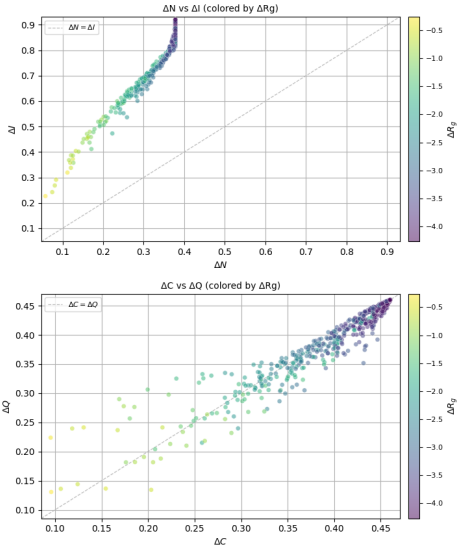
Result 4: Hierarchical Dissipation

- **Entanglement decay** is dominant but penalizes $\mathcal{R}_G$ more.
- **Classical correlations** loss (ΔC) drive ergotropy toward zero *more aggressively* than Discord (ΔQ) loss.
- LTO treats all correlations as **noise** for reservoirs, not as work fuel

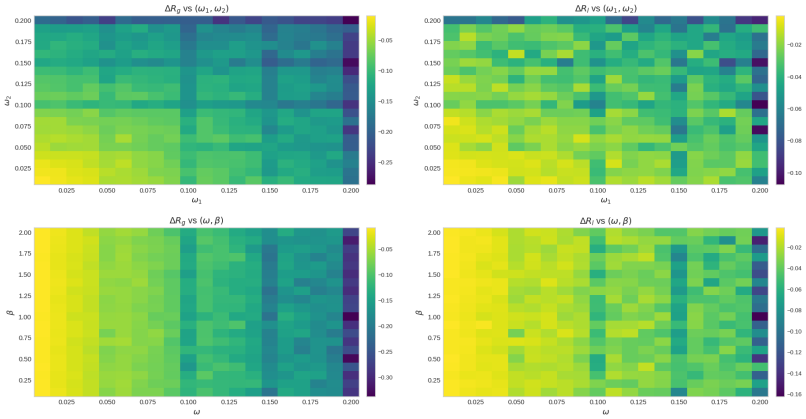
Qubit-Harmonic ($d_{\text{bath}} = 10$)

- Probed to $\omega_i, \omega_{R_i} \to 10^{-4}$ (near-zero frequency)
- High-dim bath acts as a **powerful correlation sink**
- Any local unitary remains strictly *contractive* for ergotropy

`Nelder-Mead (10 restarts, 5 min):`
Δ`global=`-3×10^{-6}, Δ`local=`-3×10^{-6}



Result 4: Harmonic-Harmonic Setup



- $\omega \in [0.01, 0.20]$ (energy level spacing)
- $\beta \in [0.1, 2.0]$ (thermal gradient)

$\Delta\mathcal{R}$ strictly ≤ 0 across *entire* phase space.

Conclusion & Future Directions

Summary of Findings

- **Definitive numerical "No-Go":** $\Delta\mathcal{R} \leq 0$ for non-degenerate case.
- Holds across qubit-qubit, qubit-harmonic, harmonic-harmonic degenerate setups.
- Degeneracy and high-dimensional baths cannot bypass monotonicity
- Quantum correlations are dissipated as **noise**, not converted to work
- Ergotropy monotone is robust to thermal gradients (β invariant)

Thank You!

Source Code:
`GitHub Repo`

Selected References:

- NumPy/SciPy/Matplotlib Docs (2026).
- Biswas et al., *Phys. Rev. E* 101, 042115 (2020).
- Alimuddin et al., *Phys. Rev. A* 99, 052320 (2019).
- Ureña et al., *Phys. Rev. Research* 6, 023023 (2024).
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Appendix: Analytical Bound Derivation

Non-Degenerate Case

LTO forces U_i diagonal in energy eigenbasis:

1. **Population invariance:** $\Phi_A(|i\rangle\langle i|) = |i\rangle\langle i|$
2. **Coherence damping:** off-diags $\to 0$
3. **Unital map:** $\lambda(\Phi(\rho)) \prec \lambda(\rho)$

By Hardy–Littlewood–Pólya theorem:

$$E_P(\Phi(\rho)) \geq E_P(\rho) \;\Rightarrow\; \Delta\mathcal{R}_{G,L} \leq 0$$

Degeneracy is a *necessary* (not sufficient) condition for any ergotropy increase.

Thermodynamic Bounds

Change in local ergotropy bounded by:

$$\Delta\mathcal{R}_L \leq \sum_i \frac{1}{\beta_i} S(\tilde{\rho}_{i,P} \| \gamma_i)$$

No correlation-dependent terms appear in the bound $\Rightarrow$ shared quantum resources *cannot* be converted to local ergotropy via LTO. Global bound:

$$\Delta\mathcal{R}_G \leq \sum_i \frac{1}{\beta_i} S(\tilde{\rho}_i \| \gamma_i) + \left[\mathrm{Tr}(H\rho_p) - \mathrm{Tr}(H|0\rangle\langle 0|)\right]$$

Appendix: Classical Correlation and Quantum Discord Computation

Classical correlations (required for Discord):

$$C(A{:}B) = S(\rho_A) - \min_{\{\Pi_i^B\}} \sum_i p_i S(\rho_{A|i})$$

Minimization over projectors $\{\Pi_i^B\}$ on the Bloch sphere (θ, ϕ) via **L-BFGS-B** with multiple random restarts for global convergence.

$$Q(A{:}B) = I(A{:}B) - C(A{:}B)$$

Harmonic Regime

Discord optimization is numerically **intractable** for harmonic oscillator systems $\Rightarrow$ mutual information I used as the primary correlation marker in that regime.

Appendix: Correlation Measures & Optimization Evidence

Nelder-Mead Runtime Output (Harmonic Setup)

Restart	$\Delta\mathcal{R}_G$	$\Delta\mathcal{R}_L$	$\Delta(\delta\mathcal{R})$
1	-2.04×10^{-4}	-3.4×10^{-5}	-1.69×10^{-4}
2	-2.04×10^{-4}	-3.4×10^{-5}	-1.69×10^{-4}
3	-2.04×10^{-4}	-3.4×10^{-5}	-1.69×10^{-4}
4	-2.04×10^{-4}	-3.4×10^{-5}	-1.69×10^{-4}
5	-2.04×10^{-4}	-3.4×10^{-5}	-1.69×10^{-4}
6	-2.04×10^{-4}	-3.4×10^{-5}	-1.69×10^{-4}
7	-2.04×10^{-4}	-3.4×10^{-5}	-1.69×10^{-4}
8	-2.04×10^{-4}	-3.4×10^{-5}	-1.69×10^{-4}
9	-2.04×10^{-4}	-3.3×10^{-5}	-1.69×10^{-4}
10	-2.04×10^{-4}	-3.3×10^{-5}	-1.69×10^{-4}

Final Result (10 restarts, ∼5h wall-clock)

Best $\Delta\mathcal{R}_G = -2.04 \times 10^{-4}$
Best $\Delta\mathcal{R}_L = -3.3 \times 10^{-5}$
Best $\Delta(\delta\mathcal{R}) = -1.69 \times 10^{-4}$

No ergotropic advantage found across all restarts.

Negativity $\mathcal{N}$

Peres–Horodecki criterion for two-qubit entanglement:

$$\mathcal{N}(\rho) = \sum_i |\lambda_i^-(\rho^{T_B})|$$

where λ_i^- are the **negative eigenvalues** of the partial transpose ρ^{T_B}. $\mathcal{N} > 0$ iff state is entangled.

Mutual Information I

Quantifies **total correlations**:

$$I(A{:}B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB})$$

where $S(\cdot) = -\mathrm{Tr}(\rho \log \rho)$ is the von Neumann entropy. Primary correlation marker in the harmonic regime where Negativity is not applicable.